

Part 1: Asset Classification

- 1. For each item below, identify the asset type, state whether it is an asset or inventory for the organization described, and write a 1–2 sentence justification. Answer in your own words — do not simply repeat the lecture definitions.**

- **A laptop computer used daily by an accountant at a CPA firm**

This laptop computer is the primary tool of productivity for the accountant at the CPA firm, making this an asset. Since it's a piece of technology that directly allows for computing function, it can be classified as an IT asset.

- **10 identical laptops sitting in sealed boxes in a Best Buy stockroom**

Since the laptops are meant to be sold by Best Buy and not used, they are inventory.

- **A Microsoft 365 E3 license assigned to one employee**

MS365 licenses are a textbook definition of an Intangible asset. Since the license is renewable and a critical component to the employee's job functions, it is an asset, not inventory.

- **A box of printer paper in the supply closet of a law firm**

Printer paper is a consumable asset when it is used; it is something that is used, runs out and needs to be replaced, but is a critical component for productivity and office work. However, when it is at rest, in the supply closet, it's not yet being used, so it is still just inventory. This one is a close one though, it almost could go either way depending on the classification convention of the specific law firm. If I had to choose one, though, I'd choose inventory since it's not at the moment being used. If it was printer paper in a printer drawer, it would be a consumable asset at that point.

- **The HVAC system in a company's server room**

The HVAC system is an Infrastructure Asset, as it specifically supports the servers and is not a general HVAC for the office.

- **A Cisco ASA 5516-X firewall protecting a business network**

An ASA 5516-X firewall is another example of an Infrastructure Asset. This one is a slightly tricky one, as it may be able to be classified as a fixed asset, but it supports the inner network for a business, so it is a critical part of the office's infrastructure.

- 2. The lecture used a laptop as an example of an item whose classification depends on context. In your own words, explain what the “context matters” principle means for asset classification. Give one original example (not used above) where the same physical item would be classified differently depending on which organization owns it. Answer in 3–5 complete sentences.**

A great example of the ‘context matters’ principle is consumable assets like Post-Its or pencils. To a real estate agency, these are consumable assets, since they support the business operations and eventually run out and need to be replaced. They are, however, considered inventory by stores like Office Depot. Interestingly, in this particular case, pencils and Post-Its can be *both* at Office Depot if employees are using them in addition to selling them. This example reflects the principle of ‘context matters’ in terms of asset classification. The item's function determines its classification.

Part 3: Depreciation Analysis

- 1. Look at the Current Book Value column (L) in your completed register and answer the following:**
 - a. Which asset currently has the lowest book value? What is it?**

Asset NETC-003 has the lowest book value. The license's current value is -\$150.00.
 - b. Which asset has depreciated the most in total dollar amount since purchase?**

Since purchase, NETC-005, the Cisco ASA 5516-X, has depreciated the most, by \$7400.
 - c. Are any assets showing a book value near \$0 or below? If so, which one(s)?**

Yes, NETC-004 (the printer) is at \$60, and NETC-002 (MS365 license) is at -\$150.
- 2. The lecture introduced the concept of “Replace vs. Refresh” as a practical use of depreciation data. Look at the Cisco ASA 5516-X Firewall in your register. Based on its current book value and the fact that it is a security-critical device, write a 3–5 sentence paragraph answering: Should this device be replaced, refreshed (upgraded), or kept as-is? Use the book value data from your register and reasoning from the lecture to support your recommendation.**

The Cisco ASA 5516-X Firewall has been in use for over six and a half years, since November of 2019. Its end of service life, or the date beyond which Cisco will not support this device, is August 31, 2026. Its current book value is \$100, which is the resale (salvage) rate of ASA 5516-X's on eBay. For all intents and purposes, this device has served out its function and will not benefit from being refreshed; because it is a critical piece of IT infrastructure and

will soon no longer be serviceable by the manufacturer, the risk from it ceasing to work is too great - it needs to be replaced.

- 3. You used straight-line depreciation for all 8 assets. However, the lecture explained that the declining balance method is often more appropriate for IT equipment because it loses value faster in early years. Identify one asset from your register where declining balance would more accurately reflect real-world value loss, and explain why in 2–4 sentences.**

The MacBook Pro 14" M3 would likely depreciate in a declining balance method in the real world. Laptops lose a lot of value up front, but then the losses are minimal as time goes on. MacBooks especially have strong resale value relative to their age and would likely reach a reasonable bottom price of somewhere around \$200 or \$300 even after a few years. The eBay listing I found for a 2024 MacBook Pro is listed for \$700, which is much closer after two years to the bottom value than the top value. The value loss per year is not static, as in straight-line depreciation, but it acts more like a curve that evens out after some time, which matches the declining balance method better than the straight-line method of depreciation.

Part 4: Audit a Broken Asset Register

Scenario: You have just been hired as a junior IT technician at a small manufacturing company. Your supervisor hands you their existing asset register and says: "We haven't touched this in two years — it's a mess. Find everything that's wrong with it before we do the physical audit next week."

Review the following asset records carefully:

Row	Asset Name	Serial #	Location	Assigned To	Purchase Date	Purchase Cost
1	Laptop	SN-48291	Office	Tom	2019	\$1200
2	PC-MFG-2019-001	DEL-7B29X-001	Floor 1, Room 104		04/01/2019	\$1,100.00
3	notebook	SN-48291	Warehouse	Luis Reyes, Ops	03/15/2022	\$989.00
4	SW-IT-2020-001	CSC-3750X-4422	Server Room	IT Dept	06/10/2020	\$4,200.00
5	Old Server	N/A	Server Room	IT Dept	2011	unknown
6	PRINTER-OFFICE	HP-LJ-9901	2nd Floor	Admin Staff	08/22/2021	\$320.00
7	PC-MFG-2019-001	DEL-7B29X-001	Floor 2, Room 208	Carla Voss, QA	04/01/2019	\$1,100.00
8	Security Cam	CAM-1190X	Lobby	Facilities	09/05/2018	\$450.00

Note: There are at least 6 distinct problems in this register. Look carefully at serial numbers, asset names, missing fields, duplicates, and assets that cannot still be in active use based on their age and description.

For each problem you find, document the following: the row(s) involved, the problem type (choose from: Ghost Asset, Bad/Inconsistent Naming, Missing Ownership, Duplicate Record, Missing/Invalid Data, or Other), what is specifically wrong, and how you would fix it. Find at least 5 problems.

Problem 1: Bad/Inconsistent Naming of assets is present rows 1, 3, 5, 6, and 8. I would rename these assets to match the official naming convention, such as CAM-SEC-2018-001 for row 8 instead of “Security Cam”.

Problem 2: Missing Ownership in row 2. I would find out who is the machine’s owner and put the correct information in the cell. Also, in row 1, “Tom” is not specific enough. He needs a last name and a department to match the rest of the table.

Problem 3: Duplicate record, rows 1 and 3. The serial numbers match, which imply that there are some big issues between these two rows – which one is right? They both have different locations, owners, prices, purchase dates, and costs. This is a huge issue for compliance and auditing, and it would be time-consuming to verify the data involved.

Problem 4: Missing data, row 5. Sure, an ‘old server’ might have missing or lost data regarding its purchase price and date, but it surely has a serial number that can be verified. A little digging into company records might reveal the purchase information, and the serial number should be an easy and quick check. This might also end up being a ghost asset if it doesn’t exist.

Problem 5: Inconsistent naming, Location column, rows 1, 6, 8. Once again, there is inconsistency in the naming of the locations. Rows 2 and 7 are good models – floor and room number. However, “Office”, “2nd Floor”, “Server Room”, and “Lobby” are not specific enough.

Problem 6 (if found): Other – inconsistencies in data formatting and style. This might be nitpicky, but it doesn’t sit well with me that row 1 is the only one whose purchase cost doesn’t contain the cents. Also, the purchase date in row 1 and row 5 only have a year, not the actual date. Also, the table would look sharper if there were no line breaks and the columns were adequately widened (I know it had to fit into a Word Doc, but hey, I did say it’s nitpicky!)

Part 5: Reflection Questions

- 1. In Lab 4, you analyzed the Equifax breach using the NIST CSF. One of Equifax's critical failures was that their IT scans failed to identify all systems running the vulnerable Struts software — in other words, they had an incomplete asset inventory. Based on what you did in this lab, describe how a well-maintained IT asset register could have helped prevent or limit the Equifax breach. Be specific about which fields in your register would have been most useful.**

A regularly maintained asset register would include important details about software like Apache Struts that would be another indicator that it would be in need of upgrading, patching, etc. Columns such as Acquisition date, Useful Life, Condition, and Warranty Expiry would show how long the software has been in place, how recent was the last patch, and what condition the software is in relative to any updates or patches. In my opinion, our asset register would benefit from more columns that would suit this scenario better: Software Version, Last Patch Date, and End of Support Date/End of Life, for example. With better asset management, Equifax may have been able to see how the software might be in need of an update soon and would have been more aware of its status. If it had been proactive, this asset register could have prevented the critical failures in the first place. At the very least, it would have shortened the investigation as it would have stuck out to internal sources that the software was outdated on the register.

- 2. The lecture's Key Risk slide warned that the most dangerous mistake in asset management is treating it as a one-time task. After building your own register in this lab, describe one specific event or workflow from real IT operations that would trigger an update to your register. Explain what field(s) would change, who would be responsible for making the update, and what could go wrong if the update was not made.**

In real IT operations, an update to the register would be required if a work laptop was transferred to a new employee. Say, for example, that an employee was getting promoted and needed a new laptop, so another employee who needed a newer machine inherited the first employee's old laptop. The fields that would need to change in the register would be Location, Assigned to, Condition, and (if the second employee was in a different department) maybe even the Asset Name. I would say that if a help desk technician was handling the migration of the laptop from the old owner to the new one, they would be responsible for updating the register. If the register was not updated, then the company might lose track of the machine as it changed hands. This could lead to an incomplete inventory and could affect insurance, auditing, internal reporting, even financials.

- 3. Imagine you are setting up asset tracking for a warehouse that holds 3,000 pieces of industrial equipment spread across four buildings. Based on the tagging methods**

covered in the lecture — barcode, QR code, RFID, and engraving — which method would you recommend, and why? What are the main trade-offs of your choice?

Considering that there are 3,000 pieces of industrial equipment over 4 warehouses, I would recommend tagging them with QR codes. Engraving each piece would take a very long time, and there wouldn't be a real need to splurge on RFID as industrial equipment is not typically moved easily or bunched together so as to scan multiple pieces at once. Bar codes and QR codes sound equally effective, except that QR codes can be scanned more conveniently by a camera on a smart phone instead of a laser scanner. For the sheer size of the asset pool as well as the convenience of having a technician scan the pieces in their proper place, I believe that using QR codes would be best.

A tradeoff of using QR codes would be that, depending on the type of equipment tagged, it may require tag replacement as wear and tear on the face of the machine might damage the QR code. Maybe an industrial grade metal print could be used to minimize wear and tear.. Additionally, it would be a massive undertaking for just one technician to scan each and every machine in four warehouses, which would be another tradeoff. Still, I believe this solution to be more effective and efficient than engraving or using RFID tags.

Sources

All salvage values and serial number examples were taken from eBay listings, warranties copied:

[Dell OptiPlex 7090](#) and [warranty](#)

[Cisco Catalyst 3650](#) and [warranty](#) (assumed [ESOL](#) as warranty expiry)

[HP Printer](#) and [warranty](#)

[ASA Firewall](#) and [warranty](#) (assumed [ESOL](#) as warranty expiry)

[MacBook Pro](#) and [warranty](#)

[APC Smart-UPS 1500VA](#) and [warranty](#) (assumed least expensive 2-year)

[Verkada CD52 Security Camera](#) (this listing has 13 cameras, so I estimated an individual camera's price based on the bulk price) and [warranty](#)